BioPathNet++ for Autoimmune Diseases: Path-Based Transfer Learning for Rare Disease Drug Repurposing—A Leakage-Controlled Study

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Abstract—Rare autoimmune diseases such as primary biliary cholangitis, systemic sclerosis, Sjögren syndrome, autoimmune hepatitis, myasthenia gravis and Crohn disease have few approved drugs and little curated data. Computational drug repurposing for them must therefore work zero-shot, ranking drugs for diseases whose treatments were never seen in training. We test whether knowledge learned from data-rich autoimmune diseases transfers to such rare ones (Aim 1), whether a path-based graph neural network beats simple repurposing rules on unseen diseases (Aim 2), and what each component contributes (Aim 3). The model, BioPathNet++, is a BioPathNet-style Neural Bellman–Ford network that runs on the full 4-million-edge PrimeKG graph on a single 16 GB GPU, with an added leakage-free transfer channel. Evaluation follows a strict protocol: each test disease is held out together with its disease–disease neighbours, and every drug is ranked against all 7,957 candidates. An audit of an earlier pipeline motivates this protocol: there, all test pairs were graph edges and a no-learning degree score reached AUROC 0.972. On 68 unseen diseases BioPathNet++ reaches an MRR of 0.394 (3 seeds, sd 0.003) and Hits@10 of 0.456, against 0.281 and 0.350 for the strongest of six baselines (paired difference +0.113, 95% CI +0.056 to +0.179, Wilcoxon $p<0.001$). On a selection-free test split of 68 further diseases an approved drug is in the top 10 of 7,957 for 76% of diseases (random ranking: 0.8%), with MRR 0.501 against 0.392 ($p=0.003$); the model outperforms every baseline evaluated on the same split. A controlled transfer curve answers Aim 1. It runs from training on rheumatoid arthritis, lupus and multiple sclerosis alone, against label-matched non-autoimmune controls, to the full graph with and without all 123 autoimmune diseases. The answer depends on how much data is available. Trained on rheumatoid arthritis, lupus and multiple sclerosis alone, the model ranks drugs for 12 held-out autoimmune diseases significantly better than when trained on label-matched non-autoimmune diseases (MRR 0.290 vs. 0.126; $p=0.016$), largely by learning which drug classes treat autoimmune disease. At full scale, however, removing all autoimmune diseases leaves rare-target accuracy unchanged (0.487 vs. 0.475). Related autoimmune diseases are the most useful knowledge when data is scarce, and the broad graph supplies it at scale. For Aim 3, path reasoning carries the accuracy, the transfer channel adds stability, the listwise loss beats BCE, and 72% of the graph's edges can be removed without loss. On TxGNN's own autoimmune split, with BioPathNet++ trained only on edges that TxGNN also trains on (three seeds each), BioPathNet++ ranks all drugs significantly better than TxGNN (MRR 0.273 vs. 0.048), and under TxGNN's own protocol neither model is significantly better (Recall@100 0.614 vs. 0.536). In case studies the model recovers ursodeoxycholic acid for primary biliary cholangitis and four cholinesterase inhibitors for myasthenia gravis. Five validation layers examine its unlabelled candidates: clinical trials, literature, drugs approved after the knowledge-graph snapshot, mechanism, and path explanations. 17% of these candidates have registered trials (random drugs: 0%; popularity baseline: 14%), and six newly approved drugs rank at a median of 1,104 of 7,957. The model is a candidate-prioritisation tool for expert review, not a prescribing tool.

Index Terms—Drug repurposing, zero-shot learning, knowledge graphs, graph neural networks, Neural Bellman–Ford networks, autoimmune diseases, rare diseases, data leakage, evaluation methodology

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I. Introduction

DRUG repurposing identifies new indications for approved drugs. Because these drugs already have known safety profiles, development is shorter and cheaper than for new molecules [1]. Rare diseases need repurposing most: of more than 7,000 rare diseases only 5–7% have an FDA-approved drug [2], and collectively they affect 3.5–5.9% of the population [3]. Autoimmune disorders are a related unmet need. In a UK cohort of 22 million people, 19 common autoimmune diseases together affected 10.2% of the population [4]. Many individual autoimmune conditions are rare and have few disease-specific drugs.

Biomedical knowledge graphs (KGs) integrate drugs, genes, proteins, phenotypes, pathways and diseases. Link prediction on these graphs can then propose drug–disease pairs [5], [6]. Most work evaluates on randomly held-out edges. Such evaluation rewards copying the drugs of near-identical diseases and does not reflect the clinical question: what could treat a disease for which nothing is known to work? The zero-shot formulation of TxGNN [2] instead holds out whole diseases. BioPathNet [7] showed that path-based reasoning with Neural Bellman–Ford networks (NBFNet) [8] further improves zero-shot indication ranking on PrimeKG.

We use transfer in this zero-shot sense: knowledge learned from diseases with known treatments is applied, without retraining, to diseases whose treatments the model has never seen. Autoimmune diseases share immune pathways, which suggests that the most useful source of that knowledge for rare autoimmune diseases could be the data-rich ones, such as rheumatoid arthritis (RA), systemic lupus erythematosus (SLE) and multiple sclerosis (MS). This paper tests that idea through three aims.

1) Aim 1, source of transfer. Must the transferred knowledge come from related, data-rich autoimmune

diseases, or does broad pharmacological knowledge suffice? We compare label-matched non-autoimmune training data and the full graph (Section VII-B).
2) Aim 2, accuracy. Does a path-based model beat simple, interpretable repurposing rules on diseases it has never seen (Section VII-A)?
3) Aim 3, components. What do path reasoning, the transfer channel, the loss and the background graph each contribute (Section VII-C)?

Answering these questions reliably required a careful protocol. Zero-shot evaluation is subtle: labels can leak into the message graph, held-out diseases can have near-duplicates in training, and random negatives make positives separable by degree alone [9], [10]. All three problems were present in an earlier version of this project (Section III). The contributions are:

- a controlled transfer curve for rare autoimmune diseases with size- and label-matched controls, which shows when autoimmune training data matters and when it does not;
- BioPathNet++, a BioPathNet-style NBFNet whose relation-wise sparse propagation keeps memory linear in the number of nodes, plus a leakage-free transfer channel, significantly better than strong baselines on unseen diseases;
- a strict-family zero-shot protocol on PrimeKG [11] with full ranking over all drugs and a selection-free test split, motivated by a leakage audit in which a degree heuristic matched a trained GNN;
- a head-to-head with TxGNN on TxGNN's own split, including a training-coverage ablation;
- a component analysis, sparsity and selection-bias analyses, and clinical-trial evidence for the top candidates.

We do not claim a new message-passing architecture. BioPathNet++ is built on BioPathNet's design, and our claims concern transfer, evaluation, efficiency and the rare-autoimmune setting.

II. Related Work

A. Knowledge graphs for drug repurposing

Hetionet integrated 29 public resources and used metapath features to prioritise repurposing candidates [5]. DRKG [12] and PrimeKG [11] followed. PrimeKG covers 17,080 diseases and adds clinical descriptions and indication, contraindication and off-label edges. Bonner et al. review these resources and show that KG embedding results depend strongly on dataset construction and evaluation choices [6], [13].

B. Embedding and GNN methods

Translational and bilinear embeddings (TransE, DistMult, RotatE) [14]–[16] and relational GCNs [17] are standard link predictors. They were used for COVID-19 repurposing [18], [19], for rare diseases from literature-derived graphs [20], for polypharmacy side effects [21], and in KG-Predict [22] and DREAMwalk [23]. A unified benchmark showed that reported KG-embedding gains often shrink under consistent training and evaluation [24]. Node-embedding methods learn a vector per node from its training edges. A disease without known drugs therefore gets little supervision, which motivates the zero-shot designs below.

C. Zero-shot and path-based methods

TxGNN [2] combines a relational GNN with a disease-similarity metric-learning module. On zero-shot disease-area splits of PrimeKG it reports relative AUPRC gains of 0.5–59.3% over baselines for indications, evaluated against sampled negatives (one per positive). NBFNet [8] represents a (source, target) pair by generalised Bellman–Ford message passing from the source. This makes it inductive over nodes, in the spirit of subgraph reasoning [25], and lets it explain predictions through paths. A*Net and ULTRA scale and generalise the idea [26], [27]. BioPathNet [7] adapts NBFNet to biomedical KGs with a background regulatory graph and stringent negative sampling. On TxGNN's zero-shot splits it reports indication MRR from about 0.2 (cardiovascular) to above 0.6 (cell proliferation, adrenal gland), and Recall@20 of 0.619 versus 0.539 for TxGNN on the cell-proliferation split. Table I positions this work.

D. Leakage in machine-learning science

Kapoor and Narayanan catalogue leakage as a main cause of irreproducible ML-based science [9]. In KG link prediction, the typical forms are test edges left in the graph, inverse or duplicate relations, and near-duplicate entities. Random negatives add a further problem: they let node popularity alone separate the classes, so AUROC becomes uninformative when positives are rare [28], [29]. Brière et al. benchmarked these effects for biomedical KG embeddings [10]. Removing train–test redundancy lowered performance, and random and cold-start splits overestimated drug-repurposing performance on an independent set of Orphanet rare-disease indications. Their permutation tests did not find that degree alone drives those models. Our audit (Section III) concerns a different failure, random negatives in evaluation, under which a degree score alone suffices.

III. Why a Strict Protocol: Audit of a Naive Pipeline

Our earlier pipeline built a graph from an enriched DRKG extract (84,712 nodes), trained an NBFNet-style model, and reported test AUPRC values between 0.935 and 0.999. We audited it with a stand-alone script that is released with the code. Table II summarises the findings.

Three mechanisms explain the high scores. (i) Target leakage: the message graph was built from the full KG, so every test pair was an edge the model could traverse. (ii) Trivial negatives: negatives were random drug–disease pairs, and indicated drugs and diseases with many indications have high degree. Degree alone therefore separates

TABLE I
Positioning against related work. Reported numbers are from the cited papers and use different splits and negatives; they are context, not a head-to-head comparison.

Method	KG	Zero-shot unit held out	Evaluation	Reported indication result
Hetionet [5]	Hetionet	none (edge-level)	metapath classifier	–
DREAMwalk [23]	multi-layer KG	none (edge-level)	association prediction	up to 16.8% accuracy gain over link-prediction models
TxGNN [2]	PrimeKG	disease area (indications, contraindications removed)	AUPRC, 1:1 sampled negatives	0.5–59.3% relative AUPRC gain over baselines across 9 areas
BioPathNet [7]	PrimeKG	TxGNN areas (+95% of molecular edges removed)	MRR, Hits@k, Recall@k over all drugs	MRR ≈0.2–0.8 by area; Recall@20 0.619 (cell proliferation)
This work	PrimeKG	disease + all disease–disease neighbours; autoimmune area	MRR, Hits@k, AUROC (all/hard/degree-matched) over all 7,957 drugs; paired tests	MRR 0.394 (68 valid.), 0.475 (6 targets), 0.449 (21 autoimmune-area diseases)

TABLE II
Audit of the earlier (invalid) pipeline.

Finding	Value
Test positive pairs present as edges of the message graph	277 / 277
Test AUROC, zero-parameter $\log \deg(\text{drug}) + \log \deg(\text{disease})$	0.972
Test AUPRC, same heuristic	0.971
Targets with out-degree 0 for the subgraph extractor (score independent of disease)	5 / 6
Drug-degree AUROC vs. other autoimmune drugs (per target)	0.50–0.84
Positive labels that are text-mined (GNBR) edges	90–91%
Spurious relation types from frequency strings	2,325

the classes, and the heuristic beat the model. (iii) One-directional subgraphs: neighbourhoods were extracted along incoming edges only, without inverse relations. Five of the six targets were then unreachable and scored by drug features alone, which is why the same drug ranked first for every disease. Further problems were text-mined labels (one of them, thalidomide for PBC, contradicts a negative randomised trial [30]), language-model features computed from identifier strings, and ablation scripts that loaded the wrong model. Every number in the rest of this paper comes from the rebuilt pipeline.

IV. Data and Zero-Shot Protocol

A. PrimeKG

We use PrimeKG [11] (Table III). Its 27 non-therapeutic relation types (protein–protein, drug–protein, disease–protein, disease–phenotype, anatomy–protein expression, drug–drug, pathways, exposures and others) form the context graph. Each relation gets an explicit inverse, giving 54 context relations. The three therapeutic relations, also with inverses, bring the total to 60. MONDO disease groupings follow PrimeKG.

Preprocessing. PrimeKG's edge file lists every edge in both directions (8,100,498 rows). For each relation we fix one canonical direction (by node type, or by node index for relations within one type), drop duplicates and only then add the inverses, so no edge enters the graph twice. Drug–disease pairs (42,631 after deduplication) are stored in a separate table and never become context edges; for each split, the message graph receives only the pairs of training diseases (Section IV). A disease family is the disease plus its disease–disease neighbours. For the targets and for RA/SLE/MS it also includes every disease node that TxGNN's name-based grouping (kg_grouped_diseases_bert_map.csv) assigns to the same group, which catches numbered subtypes. The autoimmune area re-implements TxGNN's selection exactly: all descendants of DOID:417 in the Human Disease Ontology, mapped to PrimeKG's MONDO nodes and grouped nodes through MONDO cross-references. The path channel uses no node features or text embeddings; drugs and diseases differ only by their edges. The prior features of Section V are computed on the fly from training labels and disease–protein and disease–phenotype edges. The whole build runs in about 35 s on a laptop (bpp/build_primekg.py); the 9 MB result is the dataset used in every run.

B. Labels

The model is trained to predict indications. For validation and area diseases, positives are their PrimeKG indications. For the six targets, positives are indications together with off-label uses, because some targets have almost no labelled indications: PBC 1, SSc 5, Sjögren 12, AIH 1, MG 7 and Crohn 24 drugs (50 in total). Label sparsity is intrinsic to rare diseases. The targets are therefore analysed as case studies, while quantitative claims rest on the 68 validation and 21 area diseases.

TABLE III
Dataset and split statistics.

Nodes	129,375
drugs / diseases	7,957 / 17,080
Context edges	4,007,433
relation types (with inverses)	27 (54)
Therapeutic pairs	42,631
indication	9,388
contraindication	30,675
off-label	2,568
Targets protocol	
train query diseases	1,206
validation diseases (positives)	68 (450)
target families held out	6
No-autoimmune protocol	
train / validation diseases	1,194 / 66
autoimmune nodes removed	123
area test diseases (positives)	21 (147)
Transfer: train diseases	
A / A′ / C_E	11 / 11 / 1,185
held-out autoimmune test set E	12
Test split: train / test diseases	1,065 / 68

C. Strict-family hold-out

Holding out a disease node alone is not enough. Clinical subtypes and synonyms linked by disease–disease edges often carry the same drugs: without further removal, all 16 Crohn indications remained reachable through "Crohn's colitis", MG through "adult-onset myasthenia gravis" and SSc through "scleroderma". We therefore define the family of a disease as the disease plus its disease–disease neighbours, and we hold out whole families:

- no therapeutic edge of any held-out family member is ever in the message graph;
- validation diseases are sampled at random (5% of labelled diseases, seed 0) and held out with their families, so model selection cannot reward sibling copying;
- in every training step, all therapeutic edges of the batch's query diseases and their families are also removed, so training matches the zero-shot condition at test time.

The rules are implemented in the graph builder rather than in the training script, and a unit test checks that removed edges are absent from the propagation matrices.

D. No-autoimmune protocol

To test whether accuracy on the targets comes from related autoimmune diseases, the second protocol also removes TxGNN's autoimmune disease area (DOID:417, 70 nodes) [2] and the families of all nine autoimmune diseases studied here, 123 nodes in total. Its 21 held-out area diseases with indications form an additional test set that is never used for model selection. TxGNN's area does not include PBC, SSc or Crohn disease, so the families add them.

E. Transfer conditions (Aim 1)

The transfer curve uses a fixed test set E: the 12 diseases with indications in TxGNN's autoimmune area that are neither targets nor members of the RA/SLE/MS families. E, the six targets and the validation diseases are held out with their families in every condition. Only the training diseases change:

- A: the RA/SLE/MS families (11 diseases, 161 indications), the original transfer design;
- A′: three independent sets of 11 non-autoimmune diseases, each matched to A disease by disease on number of indications (160 per set). A disease counts as non-autoimmune only if neither it nor any neighbour belongs to the 123-node autoimmune set;
- D: all 1,194 non-autoimmune diseases (the no-autoimmune protocol above);
- C_E: all 1,185 diseases except E.

The message graph only ever contains therapeutic edges of the training diseases, so each condition also defines what drug knowledge the model can see. A versus A′ isolates disease type at equal data size, and C_E versus D measures the value of autoimmune data on top of broad data. A separate "all autoimmune" condition is not possible: beyond RA/SLE/MS, PrimeKG's autoimmune diseases are the evaluation area itself. A unit test checks every split for family overlap between training and evaluation diseases.

F. Selection-free test split

Validation diseases select the stopping epoch, so their scores are slightly optimistic (Section VII-D). We therefore also draw 68 random non-autoimmune labelled diseases (seed 1) as a test set. They are held out with their families, leaving 1,065 training diseases, and are used neither for training nor for early stopping.

V. Method

Fig. 1 gives an overview.

A. Path channel

For a query (u, r_q), with disease u and query relation "indication", NBFNet [8] computes a representation of every node v that summarises all paths from u to v. Node states start from an indicator boundary condition, $h_v^{(0)} = \mathbb{1}[v = u]\, q$, where $q = \mathrm{emb}(r_q) \in \mathbb{R}^d$. Each of L layers aggregates relation-specific messages:

$$m_v^{(t)} = \frac{1}{\deg v} \sum_{(x,r,v)\in\mathcal{E}} h_x^{(t-1)} \odot w_{r_q,r}^{(t)} \; + \; \mathbb{1}[v = u]\, q, \tag{1}$$

$$h_v^{(t)} = h_v^{(t-1)} + \mathrm{ReLU}\Big(\mathrm{LN}\big(W^{(t)}\,[h_v^{(t-1)};\, m_v^{(t)};\, a_v\, m_v^{(t)}]\big)\Big). \tag{2}$$

Here $w_{r_q,r}^{(t)}$ are query-dependent DistMult relation vectors [15], $a_v = \log(\deg v + 1)/\overline{\log(\deg +1)}$ is a PNA-style degree amplifier [31], LN is layer normalisation,

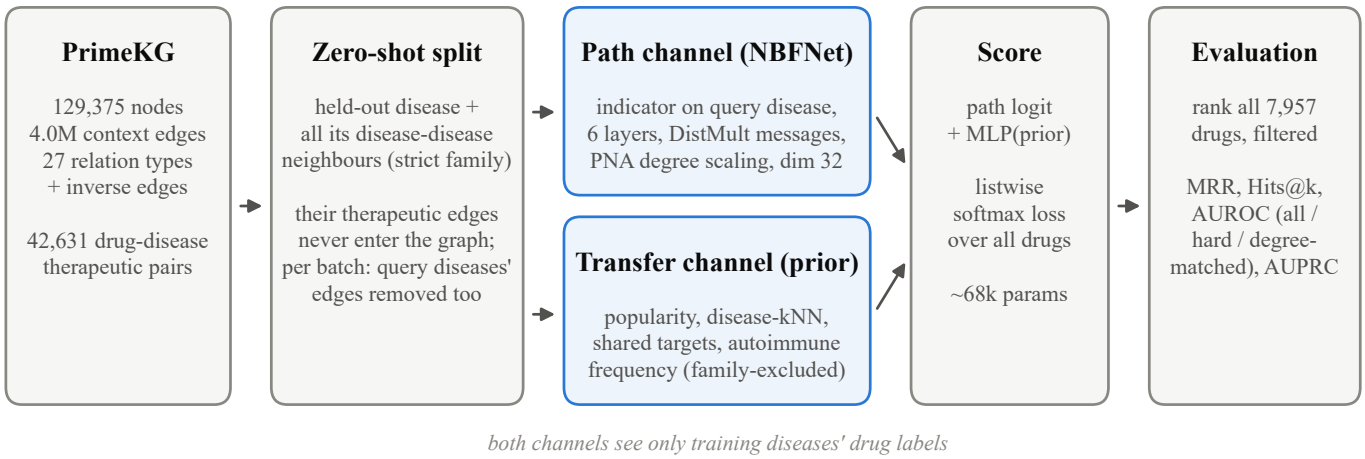


Fig. 1. BioPathNet++ pipeline. Held-out families never contribute therapeutic edges; both channels see only training diseases' labels; evaluation ranks every drug.

and the residual connection stabilises the 6-layer network. The path logit of drug d is $s_{\text{path}}(u,d) = \text{MLP}([h_d^{(1)};\ldots;h_d^{(L)};q])$. Because propagation starts at u, a single forward pass scores all 7,957 drugs. The model needs no trained embedding for u, so a disease whose drugs were never seen is handled exactly like any other node.

B. Memory-efficient full-graph propagation

NBFNet implementations usually materialise per-edge messages, which needs $O(|\mathcal{E}|\,B\,d)$ memory for B queries. That is 4 M edges × 8 queries × 32 dimensions per layer here, and it does not fit on a commodity GPU. We write (1) as $\sum_r (A_r H)\odot w_r$ over 60 sparse CSR adjacency matrices A_r. A custom autograd function stores only H and recomputes $A_r H$ in the backward pass, reducing activation memory to $O(|\mathcal{V}|\,B\,d)$. The full graph takes 0.32 GB. Training with $B = 8$ peaks at 5.8 GB and takes about 8.5 s per step on an NVIDIA T4, and the gradients pass a finite-difference check.

C. Transfer channel

The "++" component adds four prior features $\phi(u,d)$ per pair:

1) popularity, the number of training diseases indicated for d;
2) a disease-kNN transfer score, $\sum_{u'\in\mathcal{N}_k(u)} \text{sim}(u,u')\,y_{u'd}$ with $k = 20$ and cosine similarity of TF–IDF-weighted disease–protein and disease–phenotype profiles;
3) the number of d's protein targets associated with u;
4) the frequency of d among autoimmune training diseases.

Each feature is computed with u's family excluded, identically in training and testing. A two-layer MLP maps ϕ to a logit that is added to the path logit: $s(u,d) = s_{\text{path}}(u,d) + g(\phi(u,d))$.

D. Objective

Let P_u be the training indications of u. We minimise a multi-positive softmax cross-entropy over all candidates,

$$\mathcal{L}(u) = -\frac{1}{|P_u|}\sum_{d\in P_u} \log \frac{\exp s(u,d)}{\sum_{d'}\exp s(u,d')}, \tag{3}$$

so every non-indicated drug acts as a negative and no negative sampling is needed. A self-adversarial binary cross-entropy, as used for NBFNet, is evaluated as an ablation.

VI. Experimental Setup

A. Baselines

All baselines use the same protocol and the same family exclusions. Random ranking. Drug degree: context-graph degree. Popularity: number of training diseases treated. Autoimmune frequency: popularity among autoimmune training diseases. Shared targets: number of the drug's targets associated with the disease. Disease kNN: drugs of the 20 most similar training diseases, weighted by similarity. Disease kNN is a transparent version of the "guilt by association" rule used throughout repurposing [23], and the strongest baseline in our experiments.

B. Metrics

For disease u with positives P_u, each positive d is ranked among all drugs after removing the other positives (filtered rank [14]). We report

$$\text{MRR}_u = \frac{1}{|P_u|}\sum_{d\in P_u}\frac{1}{\text{rank}_u(d)}, \quad \text{H@}k_u = \frac{1}{|P_u|}\sum_{d\in P_u}\mathbb{1}[\text{rank}_u(d)\le k], \tag{4}$$

macro-averaged over diseases. MRR originates in question-answering evaluation [32] and is the standard KG-completion metric. We also report the share of diseases with at least one positive in the top 10, AUROC and

AUPRC over all drugs, AUROC against hard negatives (drugs indicated for other autoimmune diseases) and AUROC against degree-matched negatives (five drugs of nearest degree per positive). With about 7 positives among 7,957 drugs, random ranking gives MRR ≈0.001 and AUPRC ≈0.001. AUROC is insensitive to this imbalance [28], so it is not a headline metric here (Section VII-A1). For readers outside machine learning we present the metrics in order of interpretability: Any@10 first, then Hits@k, AUROC, AUPRC beside its random level, and finally MRR. MRR is easiest to read through ranks: on the test split, BioPathNet++'s MRR of 0.501 corresponds to a median known-drug rank of 2, with 43% of known drugs ranked first. A random ranking puts an approved drug in the top 10 for only 0.8% of test diseases.

C. Statistics

Models were trained with 3 seeds (0–2) where compute allowed, and one seed for the remaining ablations (stated in each table). Per-disease metrics are first averaged over seeds and then over diseases. We report seed mean ± sd, percentile bootstrap 95% CIs over diseases (1,000 resamples) [33], and, for comparisons, the paired bootstrap CI of the mean difference together with the two-sided Wilcoxon signed-rank test [34]. We call a difference significant only if both criteria agree at the 5% level.

D. Training details

Hidden size 32, 6 layers, MLP hidden size 64, dropout 0.1, Adam with learning rate 5×10^{-3}, batch of 8 query diseases (151 steps per epoch), gradient clipping at 1.0, at most 20 epochs, early stopping on validation MRR with patience 6. The full model has 68,226 parameters. These settings were fixed in advance and were not tuned. Runs used two NVIDIA T4 GPUs (16 GB each) on Kaggle, were resumable across 12-hour sessions, and stopped early after 10–20 epochs. Transfer runs train on only 11 diseases (2 steps per epoch), so they may run up to 60 epochs with patience 10; all other settings are identical.

VII. Results

We first establish accuracy on unseen diseases (Aim 2), then use the model for the transfer study (Aim 1), and finally analyse its components (Aim 3) and robustness.

A. Aim 2: accuracy on unseen diseases

Table V gives every metric for the test split, the validation diseases and the targets. On the test split, never used for training or stopping, an approved drug is in the top 10 for 76% of diseases (disease kNN 57%, popularity 19%, chance 0.8%), and AUPRC is 0.556 against a prevalence of about 0.001. Table IV and Fig. 2 compare variants. On the 68 validation diseases, BioPathNet++ reaches MRR 0.394±0.003, H@1 0.353 and H@10 0.456. Disease kNN reaches 0.281, and popularity only 0.046. The paired MRR gain over disease kNN is +0.113 (95% CI +0.056 to +0.179; Wilcoxon $p<0.001$). The model is better on 43 diseases, tied on 13 and worse on 12. The H@10 gain is also significant (+0.105, $p=0.002$). For 56% of unseen diseases at least one approved drug appears in the top 10 of 7,957, compared with 44% for disease kNN and 15% for popularity. Against hard negatives (drugs of other autoimmune diseases), AUROC is 0.799, versus 0.661 for disease kNN and 0.476 (below chance) for popularity. The model therefore does more than rank generically popular drugs. It outperforms every baseline evaluated on the same leakage-free split. BioPathNet was not run on this split; TxGNN is compared on its own split in Section VII-D2.

On the six targets MRR is 0.475±0.017 versus 0.296. The difference (+0.179, CI -0.027 to +0.370) points the same way on 4 of 6 targets. With six diseases it is not significant (Wilcoxon $p=0.188$), and we do not claim it is.

1) Why AUROC is not the headline metric: Fig. 3 plots AUROC against MRR for every method. Baselines without learning reach AUROC 0.78–0.86, and popularity reaches 0.830, yet their MRR is below 0.06. With thousands of negatives per positive, AUROC rewards placing positives above the long tail of irrelevant drugs. It says little about whether a positive reaches the top of the list, which is what a clinician would read. The same effect let a degree heuristic reach AUROC 0.972 in the audit. We therefore headline MRR and Hits@k and report AUROC for completeness (0.938 validation, 0.982 targets).

B. Aim 1: transfer to rare autoimmune diseases

Fig. 4 and Table VI give the transfer curve. Scarce data. Trained only on the RA/SLE/MS families (A), BioPathNet++ reaches MRR 0.290 on E, against 0.126 when trained on label-matched non-autoimmune diseases (A′). The paired difference of +0.164 (CI +0.037 to +0.316; Wilcoxon $p=0.016$; better on 9 of 12 diseases) is significant. It also holds when each run's final checkpoint is used instead of the early-stopped one (+0.146, CI +0.046 to +0.263, $p=0.021$). The rare targets point the same way (0.124 vs. 0.041; $p=0.062$, $n=6$). The disease-kNN baseline shows the same pattern (0.258 vs. 0.086), and in this regime the model is only slightly ahead of it. With 11 training diseases, the transferable knowledge is essentially a drug-class prior. The RA/SLE/MS families are treated mostly with corticosteroids and TNF inhibitors, and the A-trained model ranks the same corticosteroids first for every target. That is a reasonable first guess for an autoimmune disease, but it is not disease-specific reasoning. To test whether path reasoning itself transfers in this regime, rather than only the drug-class prior carried by the transfer channel, we repeated A and A′ with paths only (3 seeds for A, one per control draw). Both stay near the floor (random ranking: 0.001): 0.007 for A against 0.010 for A′. A is even marginally lower (difference -0.003, CI -0.005 to -0.001; $p=0.034$; better on 3 of 12 diseases), a difference too small to matter. Path reasoning alone thus learns almost nothing from 11 training diseases; the scarce-data advantage of A comes from the transfer channel.

TABLE IV
Zero-shot drug ranking over all 7,957 drugs (targets protocol). Models: mean±sd over seeds (n). H@k: fraction of a disease's known drugs in the top k. $\text{AUROC}_{\text{hard}}$: known drugs vs. drugs of other autoimmune diseases.

	68 zero-shot validation diseases					6 rare autoin	
Method	MRR	H@1	H@10	AUROC	$\text{AUROC}_{\text{hard}}$	MRR	H@
Random	0.001	0.000	0.002	0.471	0.488	0.001	0.0
Drug degree	0.004	0.000	0.003	0.783	0.421	0.003	0.0
Popularity	0.046	0.034	0.049	0.830	0.476	0.099	0.1
Autoimmune frequency	0.038	0.032	0.040	0.827	0.128	0.094	0.1
Shared targets	0.057	0.032	0.105	0.845	0.530	0.008	0.0
Disease kNN	0.281	0.239	0.350	0.860	0.661	0.296	0.4
Prior only (transfer channel) (n=1)	0.303	0.265	0.353	0.884	0.707	0.322	0.4
Paths only (NBFNet) (n=3)	0.405±0.013	0.363±0.019	0.479±0.025	0.946±0.004	0.829±0.011	0.463±0.087	0.657±
BioPathNet++ (paths + prior) (n=3)	0.394±0.003	0.353±0.008	0.456±0.018	0.938±0.001	0.799±0.010	0.475±0.017	0.648±

TABLE V
All metrics on the selection-free test split (3 seeds), the validation diseases and the rare targets (seed means). Any@10: diseases with an approved drug in the top 10 (random expectation 0.8%). AUPRC of a random ranking ≈ prevalence (≈0.001). Mean rank among 7,957 drugs.

Set	Method	Any@10	H@1	H@10	H@50	AUROC	$\text{AUROC}_{\text{hard}}$	$\text{AUROC}_{\text{deg}}$	AUPRC	MRR	Mean rank
Test (68, never selected)	Random	1%	0.000	0.005	0.006	0.453	0.471	0.500	0.002	0.001	4,349
	Popularity	19%	0.042	0.070	0.267	0.885	0.562	0.812	0.084	0.062	918
	Disease kNN	57%	0.347	0.459	0.583	0.907	0.721	0.859	0.429	0.392	742
	BioPathNet++	76%	0.435	0.637	0.727	0.959	0.862	0.924	0.556	0.501	329
Validation (68)	Random	1%	0.000	0.002	0.004	0.471	0.488	0.490	0.002	0.001	4,205
	Popularity	15%	0.034	0.049	0.188	0.830	0.476	0.760	0.069	0.046	1,355
	Disease kNN	44%	0.239	0.350	0.490	0.860	0.661	0.822	0.313	0.281	1,118
	BioPathNet++	56%	0.353	0.456	0.601	0.938	0.799	0.886	0.424	0.394	495
Rare targets (6)	Random	0%	0.000	0.000	0.000	0.476	0.488	0.389	0.002	0.001	4,165
	Popularity	50%	0.062	0.148	0.162	0.905	0.460	0.766	0.116	0.099	755
	Disease kNN	67%	0.243	0.424	0.725	0.962	0.778	0.917	0.407	0.296	299
	BioPathNet++	83%	0.396	0.648	0.695	0.982	0.892	0.954	0.594	0.475	141

TABLE VI
Transfer curve: MRR [95% CI] by training condition. Test diseases are held out in every condition.

		12 held-out autoimmune-area diseases (E)		6 rare targets	
Training data	#dis.	BioPathNet++	disease kNN	BioPathNet++	disease kNN
A′: 11 matched non-autoimmune diseases	11	0.126 [0.04, 0.22]	0.086 [0.02, 0.17]	0.041 [0.00, 0.08]	0.011 [0.00, 0.02]
A: RA/SLE/MS families (11 diseases)	11	0.290 [0.09, 0.50]	0.258 [0.06, 0.47]	0.124 [0.01, 0.26]	0.120 [0.01, 0.26]
D: all non-autoimmune diseases	1,194	0.532 [0.34, 0.72]	0.318 [0.13, 0.53]	0.487 [0.20, 0.73]	0.273 [0.08, 0.56]
C_E: all diseases except the test set	1,185	0.452 [0.25, 0.67]	0.271 [0.08, 0.50]	0.550 [0.28, 0.78]	0.299 [0.07, 0.57]

Full scale. Broad training data without any autoimmune disease (D) raises accuracy on E to 0.532. Adding the autoimmune diseases (C_E, one seed) gives 0.452 (difference -0.081, p=0.232), with no measurable gain. On the rare targets, removing all autoimmune diseases likewise leaves MRR unchanged (0.487 vs. 0.475; 3 seeds each; p=0.812). Aim 1 is therefore answered in two parts. The model's zero-shot transfer works at every scale we tested (Aim 2). Its best source depends on how much data is available: related autoimmune diseases when training data is scarce, and the broad pharmacological graph, which then makes autoimmune data unnecessary, at full scale. One caveat concerns model selection. The A and A′ runs early-stop on the 68 general validation diseases, on which all of them score MRR below 0.05, so the selected epoch can be very early (epoch 0 for one A seed). Training on 11 diseases is also unstable: one A seed falls from 0.29 to 0.09 MRR on E by its final epoch. The final-checkpoint analysis above shows that the A versus A′ conclusion does not depend on this choice.

TABLE VII
Effect of removing all 123 autoimmune diseases from training.

Training data	6 targets		21 autoimmune-area diseases	
	MRR	H@10	MRR	H@10
BioPathNet++, all diseases in training (n=3)	0.475±0.017	0.648±0.033	–	–
BioPathNet++, 123 autoimmune diseases removed (n=3)	0.487±0.031	0.615±0.022	0.449±0.042	0.640±0.037
Paths only, 123 autoimmune diseases removed (n=1)	0.480	0.642	0.393	0.617
Disease kNN (no-autoimmune pool)	0.273	0.594	0.311	0.469

1) Transfer at full scale: Table VII repeats the question at full scale on all 21 area diseases. With all 123 autoimmune nodes removed from training, target MRR is 0.487±0.031 versus 0.475±0.017 with them (3 seeds each;

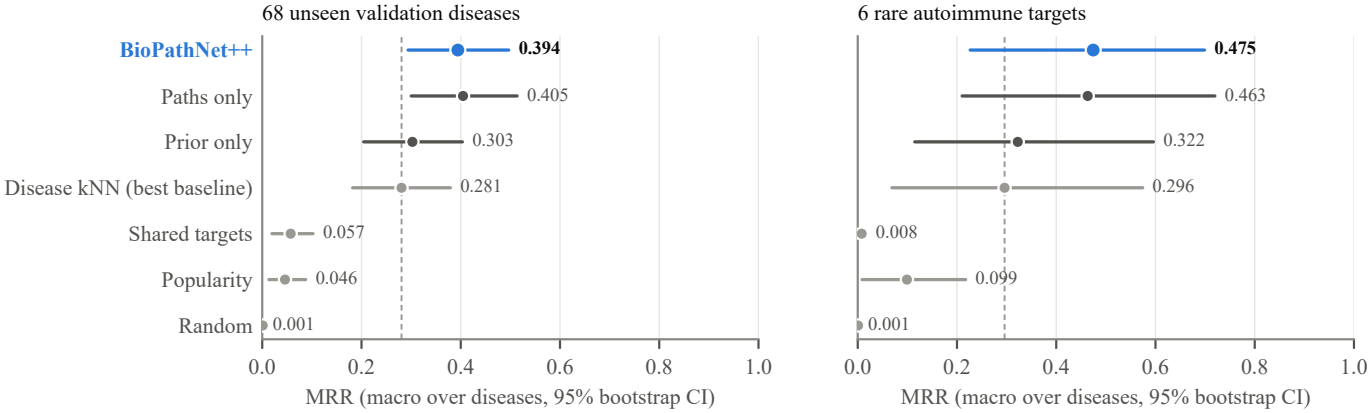


Fig. 2. MRR with 95% bootstrap CIs over diseases (seed-averaged). Dashed line: disease kNN, the strongest baseline.

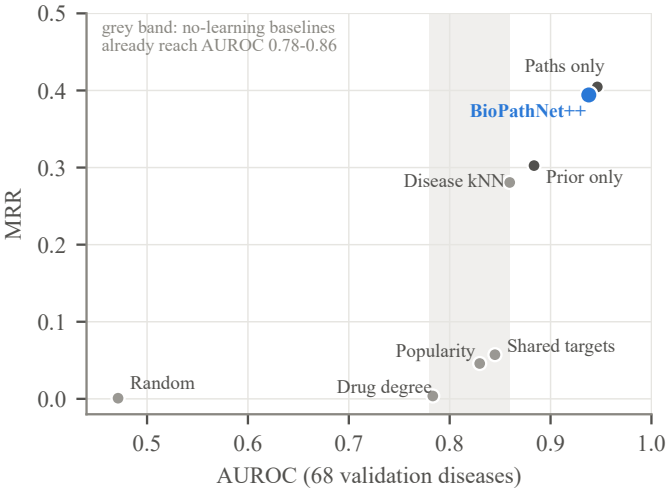


Fig. 3. AUROC vs. MRR on the 68 validation diseases. AUROC is high even for no-learning baselines; MRR separates methods.

paired difference with minus without -0.013, CI -0.070 to +0.055, $p=0.812$). Removing autoimmune training data does not hurt the autoimmune targets. On the 21 held-out autoimmune-area diseases, which are never used for selection, the model reaches MRR 0.449±0.042 versus 0.311 for disease kNN. The difference is +0.138 (CI +0.010 to +0.280), but the Wilcoxon test gives $p=0.147$, so by our criterion it is suggestive rather than significant. The autoimmune-specific transfer hypothesis is therefore not supported. Whatever transfers comes from the general pharmacology and biology encoded in the KG, not from autoimmune training labels.

In this harder setting the transfer channel may help. On the area diseases, paths only (one seed) reaches 0.393 against the full model's 0.449 (difference +0.056, CI +0.009 to +0.110, $p=0.073$). This is not significant, and more paths-only seeds are needed before claiming it.

C. Aim 3: what each component contributes

Table VIII isolates the components. Without the transfer channel (paths only), validation MRR is 0.405, statistically tied with the full model (difference -0.010, $p=0.148$).

TABLE VIII
Ablations (targets protocol). p: two-sided Wilcoxon on paired validation MRR vs. the full model.

Variant (seed 0 unless noted)	Valid MRR	Valid H@10	Targets MRR	p vs full (valid)
Full model (n=3)	0.394±0.003	0.456±0.018	0.475±0.017	–
− transfer channel (n=3)	0.405±0.013	0.479±0.025	0.463±0.087	0.148
− path reasoning (prior only)	0.303	0.353	0.322	<0.001
BRG relation subset	0.397	0.488	0.424	0.906
BCE loss instead of listwise	0.347	0.434	0.292	0.005

Without path reasoning (transfer channel only), MRR falls to 0.303 ($p<0.001$), close to disease kNN, which is one of its inputs. Path reasoning is therefore the source of the gain. The transfer channel does not change the mean. It does reduce variability across seeds on the targets, from sd 0.087 to 0.017 (Fig. 6), which matters in practice because a ranking used for case studies should not depend on the seed. An earlier interim claim that the channel improved rare-target accuracy, based on one seed, did not survive replication and was withdrawn.

Two further single-seed ablations change one design choice each. Loss. Replacing the listwise softmax with NBFNet's self-adversarial BCE lowers validation MRR to 0.347. Against the seed-averaged full model the gap is +0.047 (Wilcoxon $p=0.005$). Against the same seed alone it is +0.048 (CI -0.001 to +0.100; $p=0.023$), so the advantage is consistent but borderline for one seed. Scoring all drugs in the loss matches how the model is evaluated, which is a plausible reason for the gain. Background graph. Following BioPathNet's background regulatory graph [7], we drop the two largest and least mechanistic relation families: tissue expression and drug–drug interactions. This shrinks the context graph from 4,007,433 to 1,133,029 edges (72% fewer). Validation MRR is unchanged (0.397; $p=0.906$), and steps are 11% faster (7.49 vs. 8.46 s) with lower peak memory (5.52 vs. 5.76 GB). Most of PrimeKG's edges are therefore redundant for this task, in line with BioPathNet's relation ablation [7], and a 3.5× smaller graph is a practical option for limited hardware.

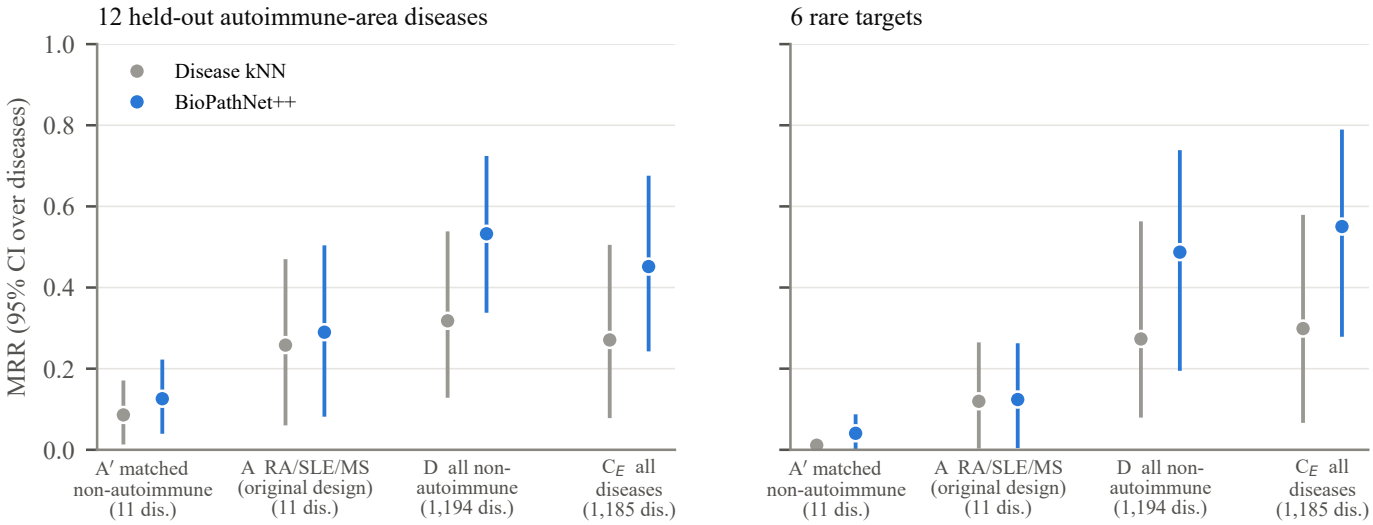


Fig. 4. Transfer curve (Aim 1). MRR on the 12 held-out autoimmune diseases E (left) and the six rare targets (right) for each training condition, BioPathNet++ versus disease kNN, with 95% bootstrap CIs over diseases. A′ averages three matched control sets.

TABLE IX
Selection-free test split: 68 diseases never used for training or early stopping.

Method (68 test diseases, never used for training or stopping)	MRR	H@10	$\text{AUROC}_{\text{hard}}$
Random	0.001	0.005	0.471
Popularity	0.062	0.070	0.562
Disease kNN	0.392	0.459	0.721
BioPathNet++	0.501	0.637	0.862

TABLE X
TxGNN's autoimmune disease-area split, generated by TxGNN's own code (13 test diseases). Main comparison (full coverage): BioPathNet++ keeps every non-area disease in training and uses only edges that TxGNN also trains on; means over seeds. Ablation: our first run, in which BioPathNet++ was trained only on diseases without any TxGNN test edge. TxGNN protocol: only drugs with at least one indication are ranked; full ranking: all 7,957 drugs. The published row is from TxGNN's repository (different random draw of the split).

		TxGNN protocol			full ranking		
Method	seeds	R@100	AUROC	AUPRC	MRR	H@100	AUROC
TxGNN, published	5	0.338	–	–	–	–	–
Full training coverage (main comparison)							
TxGNN	3	0.536	0.901	0.161	0.048	0.232	0.753
BioPathNet++ (1,139 train dis.)	3	0.614	0.777	0.343	0.273	0.618	0.936
Strict coverage: every disease with a test edge held out (ablation)							
TxGNN (unchanged)	1	0.474	0.909	0.147	0.026	0.155	0.736
BioPathNet++ (637 train dis.)	1	0.475	0.746	0.101	0.061	0.398	0.910

D. Robustness: sparsity, selection bias and a selection-free test

Validation diseases also select the stopping epoch, so validation scores are optimistic. From the training logs, the gap between the best-epoch and last-epoch validation MRR is 0.002–0.016 for the full model and 0.022–0.030 for paths only. This is an order of magnitude smaller than the gain over disease kNN (+0.113), but larger than the difference between the full model and paths only, so the latter is a tie. The targets and the 21 area diseases are untouched by selection, and their results agree with the validation conclusions.

1) Selection-free test split: On the selection-free test split (Table IX), BioPathNet++ reaches MRR 0.501 and H@10 0.637, against 0.392 and 0.459 for disease kNN (paired difference +0.109, CI +0.044 to +0.180, p=0.003; 3 seeds). For 76% of the test diseases an approved drug appears in the top 10 (disease kNN: 57%). The same model reaches 0.378 on its validation diseases. Test diseases are drawn from non-autoimmune diseases and are easier for every method (disease kNN also rises from 0.281 to 0.392), so the two levels are not directly comparable. What matters is that the gain over the baseline holds on diseases that no model choice has seen.

2) Head-to-head with TxGNN on its own split: For a direct comparison we ran both models on TxGNN's own autoimmune disease-area split, generated by TxGNN's code (commit 0e0203c). Its random test edges are drawn without a seed, so we fixed the seed; this draw therefore differs from the published runs. The test set is the 13 held-out area diseases that have indications (75 drug–disease pairs). Besides all drug edges of the area diseases, TxGNN's test set contains 5% of edges drawn at random from the area's two-hop neighbourhood, so 851 other diseases also lose some drug edges to it. We converted the split to our format so that our model never sees an edge that TxGNN does not train on. These 3,248 random test edges are removed from both our message graph and our training labels, and the area diseases are held out with their disease–disease families, as in our protocol. BioPathNet++ thus trains on 1,139 diseases and uses 63 more for early stopping, whereas TxGNN learns from the training edges of all 1,299 non-test diseases with indications. Our model still sees less than TxGNN. Our protocol also holds out 64 relatives of test or validation diseases, and our message graph contains drug–disease edges only of training diseases, so it lacks, for example, the 10,206 contraindication and off-label edges of 667

diseases without any indication. These differences work against BioPathNet++. TxGNN was trained with the settings of its reproduction script: one full pre-training epoch (1.24 h on a T4) and 500 fine-tuning epochs. Each model was trained with 3 seeds on the same split; we report means over seeds and run the paired tests on per-disease seed means. As a check of the evaluation code, our implementation of TxGNN's protocol reproduces TxGNN's own evaluation of its predictions exactly for every seed (mean Recall@100 0.536, higher than the published 0.338 for a different draw of the split).

With full training coverage (Table X), BioPathNet++ ranks the full drug space much better. Over all 7,957 drugs its MRR is 0.273 (sd 0.009 over seeds) against 0.048 (sd 0.049) for TxGNN (difference +0.226, CI +0.055 to +0.422, p=0.040). H@100 is 0.618 against 0.232 (+0.386, CI +0.197 to +0.589, p=0.002), and AUROC is 0.936 against 0.753 (+0.183, CI +0.094 to +0.290, p<0.001; better on 13 of 13 diseases). These differences, and the one in H@10 (+0.365, p=0.027), are significant by our rule. Under TxGNN's own protocol, which ranks only the 1,798 drugs with at least one indication, no difference is significant. BioPathNet++ has the higher Recall@100 (0.614 against 0.536; +0.078, CI -0.090 to +0.246, p=0.389) and AUPRC (0.343 against 0.161; +0.182, CI -0.003 to +0.395, p=0.168). TxGNN has the higher AUROC (0.901 against 0.777; -0.124, CI -0.259 to -0.010, p=0.168). Its confidence interval excludes zero but the Wilcoxon test does not reject, and TxGNN is better on 7 of 13 diseases, so by our rule this advantage is not significant. TxGNN separates the true drugs from other indicated drugs more evenly across the list, while the higher AUPRC and Recall@100 of BioPathNet++ point to more true drugs near the top. The ranked lists explain why the two protocols differ: only 23% of TxGNN's top-100 drugs have an indication for any disease in the graph, against 99% for BioPathNet++. TxGNN ranks many drugs without any indication near the top, and its protocol removes them before scoring. TxGNN's full-ranking MRR also varies strongly between seeds.

Coverage ablation. Our first run used a stricter conversion. To keep every TxGNN test edge out of our graph, it held out every disease with any test edge, so BioPathNet++ trained on only 637 diseases (one seed per model; lower block of Table X). In that run Recall@100 was tied (0.475 against 0.474), TxGNN's AUROC advantage was significant (0.909 against 0.746; CI -0.245 to -0.080, p=0.003), and the MRR gain over all drugs was not (0.061 against 0.026; CI -0.034 to +0.105). TxGNN's training data is the same in both runs. Full coverage raised our model's full-ranking MRR by +0.213 (CI +0.073 to +0.391, p=0.013), its H@100 by +0.220 (p=0.016) and its AUPRC under TxGNN's protocol by +0.242 (p=0.013). Both of its AUROCs changed little (+0.031, p=0.376; +0.026, p=0.273). TxGNN's AUROC advantage under its own protocol shrank only from -0.163 to -0.124; it is no longer significant mainly because it is less consistent across diseases (TxGNN better on 7 rather than 10 of 13). The strict conversion therefore mainly understated how well BioPathNet++ places true drugs near the top of the list. The ablation compares a three-seed mean with a single seed, so it contains some seed noise, although BioPathNet++'s seeds differ little (MRR sd 0.009).

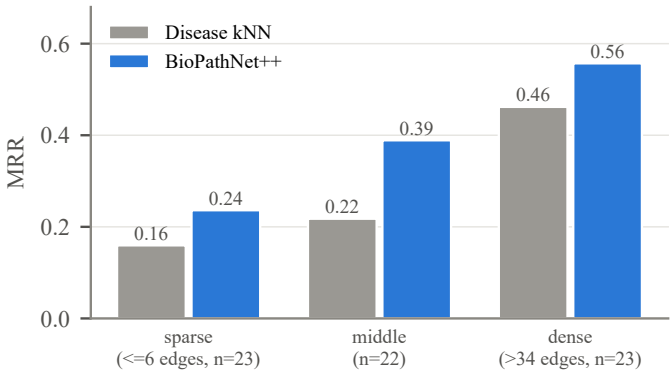


Fig. 5. Validation MRR by connectivity tertile of the held-out disease (number of context edges).

TABLE XI
Per-target MRR (seed mean) and the model's top predictions (seed 0). * labelled in PrimeKG.

Target	#known	kNN	Paths only	BioPathNet++	Best rank	Top-3 predictions (seed 0)
PBC	1	1.000	1.000	1.000	1	Ursodeoxycholic acid*, Doripenem, Cholic Acid
SSc	5	0.041	0.437	0.452	1	Benzoic acid*, Bosentan*, Limaprost
Sjögren	12	0.152	0.731	0.510	1	Prednisolone*, Polyethylene glycol 400*, Propylene glycol*
AIH	1	0.001	0.002	0.002	>100	Prednisolone, Triamcinolone, Hydrocortisone
MG	7	0.179	0.418	0.671	1	Ambenonium*, Neostigmine*, Pyridostigmine*
Crohn	24	0.403	0.191	0.212	1	Hydrocortisone*, Triamcinolone*, Cortisone acetate*

Trained on a subset of TxGNN's training edges, BioPathNet++ ranks the full drug space significantly better than TxGNN and is not significantly worse under TxGNN's own protocol. The comparison still rests on one split and 13 test diseases from one disease area.

3) Sparsity: Rare diseases are sparsely annotated, so we stratify the validation diseases by their number of context edges (Fig. 5). Accuracy grows with connectivity (Spearman ρ = 0.30, p=0.012). BioPathNet++ is ahead of disease kNN in every tertile: 0.236 vs. 0.160 for the sparsest third, 0.389 vs. 0.218 in the middle, and 0.557 vs. 0.462 for the densest third. The relative gain is largest in the middle tertile. Diseases with almost no molecular annotation remain the hardest case for every method.

E. Case studies

Table XI and Fig. 6 show the six targets. PBC. Ursodeoxycholic acid, the first-line therapy and the only labelled drug, is ranked first in all seeds. Cholic acid and dehydrocholic acid, which are also bile acids, follow. MG. The four labelled cholinesterase inhibitors (ambenonium, neostigmine, pyridostigmine, distigmine) are ranked 1–4 in all seeds. SSc. The labelled drugs aminobenzoic acid, benzoic acid and bosentan lead, followed by corticosteroids. Sjögren. Labelled tear-substitute components (propylene glycol, povidone, polysorbate 80) and prednisolone lead. Crohn. Labelled corticosteroids hold ranks 1–4. Mesalazine (off-label) and fluoroquinolones follow, but many of the 24 labelled drugs rank low, which keeps MRR modest. AIH.

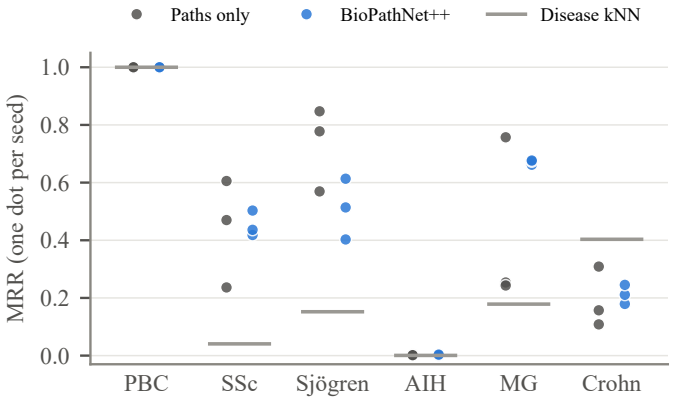


Fig. 6. Per-target MRR for each seed. The transfer channel narrows the spread across seeds (SSc, Sjögren, MG).

TABLE XII
Evidence for unlabelled candidates: BioPathNet++ top-20 vs. popularity top-20 vs. random drugs (six targets).

Evidence for unlabelled candidates	BioPathNet++ top-20 (n=86)	Popularity top-20 (n=120)	Random (n=120)
Trial names the drug, grade A–C	17%	14%	0%
Trial names the drug, phase ≥2 (A–B)	15%	12%	0%
Trial tests the drug, grade A–C	13%	8%	0%
Trial tests the drug, phase ≥2 (A–B)	12%	8%	0%
Any PubMed co-mention	63%	72%	18%
PubMed: evidence of benefit	15%	10%	0%
PubMed: harm or no benefit	5%	12%	0%
≥1 drug target among disease genes	17%	16%	3%

The model fails by the metric but arguably not clinically. PrimeKG lists a single off-label drug for AIH, while the model ranks prednisolone first in two of three seeds. Corticosteroids are standard AIH therapy and the comparator arm in AIH trials (prednisone in NCT00838214). This is a label gap, and it shows that metric values on sparsely labelled diseases are lower bounds.

F. Validation of the predictions

Ranking metrics only reward drugs that PrimeKG already labels. To judge the model's unlabelled candidates, we built five further validation layers (bpp/validate.py; every API response is cached with the code). The candidate set is each target's top 20 by rank averaged over the three seeds; 86 of these 120 are unlabelled. We compare them with two reference sets: the popularity baseline's top 20 unlabelled drugs and 20 random unlabelled drugs per target (Table XII).

Clinical trials. We searched ClinicalTrials.gov (API v2) for each drug and disease, with disease synonyms. A trial counts only if it names the drug as an intervention, matched by the whole drug name (so prednisolone does not match methylprednisolone). Grades: A, phase ≥3; B, phase 2; C, phase 1 or observational; D, none. 17% of the model's unlabelled candidates have such a trial (11 at phase ≥3), against 14% for popularity and 0% for random drugs. Naming the drug is not the same as testing it: corticosteroids often appear as background therapy, premedication or transplant conditioning. A stricter count therefore keeps only trials whose title names the drug or in which the drug is in some but not all arms. Under this rule, 13% of the model's candidates have a trial (9 at phase ≥3), against 8% for popularity and 0% for random drugs (Table XII). Examples are amifampridine for MG (NCT03304054), ciprofloxacin (NCT00609973) and clarithromycin (NCT00269386) for Crohn, N-acetylcysteine for SSc (NCT00428883), and ustekinumab for Crohn, which the FDA approved for Crohn in 2016 but PrimeKG does not label. We checked every trial cited here on ClinicalTrials.gov by hand. A trial is not evidence of benefit: amifampridine helped in a phase-2b trial in MuSK-antibody MG [35], but in a 2026 crossover trial in acetylcholine-receptor-antibody MG it was not better than pyridostigmine alone and caused more adverse events [36]. Some grades for PBC and AIH come from PBC–AIH overlap-syndrome trials.

The same check on the C_E transfer model's top 10 for the 12 held-out autoimmune-area diseases finds an approved drug in the top 10 for 9 of 12 diseases. 21 of its 74 unlabelled top-10 drugs have a registered trial for the disease, among them canakinumab for CINCA syndrome, which has phase 3 trials (NCT00685373) (transfer_E_top10.csv).

Literature. For each pair we retrieved the PubMed hit count for "drug"[tiab] AND "disease"[tiab] and the three most relevant titles. The titles were classified as supporting, mixed, against or unrelated with AI assistance (literature_labels.csv). We then re-read the abstracts behind every supporting and against label. Studies in animals only, drugs given in every arm and inconclusive results count as mixed, and negative trials count as against. This changed 17 of the labels; most of the changes moved a pair from supporting to mixed. The model's candidates are co-mentioned with their disease somewhat less often than popular drugs (63% vs. 72%), but more often in supporting papers (15% vs. 10%; random 0%). They are also less often in papers that report harm or no benefit (5% vs. 12%). The model's 4 such pairs are three corticosteroid pairs and N-acetylcysteine for SSc. The steroid pairs are the renal-crisis risk in SSc, a negative prednisone trial in Sjögren syndrome and one case of liver injury; N-acetylcysteine was negative in SSc digital-ulcer trials. Popularity's are mostly antibiotics that can worsen MG or induce autoimmune hepatitis.

Temporal validation. PrimeKG's drug labels predate 2022. Of the drugs approved for our targets since then, six are drug nodes in PrimeKG (Table XIII); seladelpar, efgartigimod, zilucoplan, nipocalimab and mirikizumab are absent. Their median rank is 1,104 of 7,957 (top 14%), and 5 of 6 rank better than the random median. Risankizumab, the first IL-23 inhibitor approved for Crohn (2022), ranks 37. Elafibranor, a PPAR agonist for PBC, ranks in the bottom third. With six drugs the trend is not significant (sign test p=0.109), and disease kNN ranks three of the six higher. Three older approvals that PrimeKG fails to label are reported separately: obeticholic acid, nintedanib and tocilizumab. Tocilizumab for SSc-associated lung disease ranks 58.

Mechanism. Using PrimeKG's drug–target and disease–gene edges, 17% of the model's unlabelled candidates

TABLE XIII
Ranks of drugs approved for a target after the PrimeKG snapshot (top block) and of earlier approvals missing from PrimeKG's labels (bottom). Mean rank over three seeds; dates from the manufacturers' or FDA announcements.

Drug	Target	FDA approval	BioPathNet++ rank	Top %	*k*NN rank	Popularity rank
Risankizumab	Crohn	2022-06	37	0.5	30	951
Upadacitinib	Crohn	2023-05	550	6.9	127	1,527
Ravulizumab	MG	2022-04	845	10.6	1,383	1,381
Guselkumab	Crohn	2025-03	1,363	17.1	5,570	5,570
Rozanolixizumab	MG	2023-06	2,278	28.6	7,182	7,182
Elafibranor	PBC	2024-06	5,825	73.2	3,266	3,266
Tocilizumab	SSc	2021-03	58	0.7	35	260
Nintedanib	SSc	2019-09	848	10.7	375	365
Obeticholic acid	PBC	2016-05	2,272	28.6	6,452	6,452

have at least one target among the disease's genes, against 16% for popularity and 3% for random drugs. A hypergeometric test for enrichment of drug targets in 109 immune-related Reactome pathways (Benjamini–Hochberg corrected) found no significant enrichment in any set. Most candidates have few annotated targets, so this test has little power.

Path explanations. We extract the paths behind a prediction by gradient-times-input attribution on every edge's message, followed by beam search over the most important edges [8] (bpp/explain.py). A check confirms that the attribution pass reproduces the model's scores exactly. Table XIV shows four examples. The cholinesterase inhibitors reach MG through the MuSK gene and neuromuscular-junction disease, and amifampridine through its indication for Lambert–Eaton myasthenic syndrome. Ursodeoxycholic acid reaches PBC through biliary-tract diseases. The paths also expose weak predictions: ciprofloxacin reaches Crohn through "mouth disease" and sinusitis, not through its documented use in perianal disease.

Drug classes and popularity bias. By name, corticosteroids fill 41% and antibacterials 21% of all 120 top-20 slots. DRKG's DrugBank-to-ATC links cover only 50% of the candidates, so we use the drug names as the primary class signal and report ATC codes alongside. As an exploratory complement we re-rank each target after regressing out log popularity. The resulting "less obvious" lists include amifampridine and guanidine for MG, lifitegrast for Sjögren and the IL-23 inhibitor tildrakizumab for Crohn (popularity_adjusted_candidates.csv).

Expert review. The top 10 per target, with all evidence columns and an empty 1–5 plausibility column, are prepared for clinical review (expert_review_sheet.xlsx).

VIII. Discussion

A. How good are these numbers?

Two comparisons matter. Against chance and simple rules, the model improves MRR from 0.001 (random) and 0.046 (popularity) to 0.394, and it is significantly better than a strong, interpretable similarity baseline. Against the literature, BioPathNet reports zero-shot indication MRR from about 0.2 to above 0.6 across disease areas, and Hits@10 from 30% to 100% [7]. Our values fall inside that range. The protocols differ, however. BioPathNet and TxGNN hold out disease areas and remove 95% of their molecular edges, whereas we hold out families and keep molecular context. TxGNN's AUPRC uses one sampled negative per positive, while our AUPRC uses all drugs. We therefore do not compare with their reported numbers, and compare with TxGNN only on a shared split (its own; Section VII-D2). BioPathNet has not yet been run on a shared split.

B. Why the earlier numbers looked better

The earlier pipeline reported AUPRC up to 0.999 because it measured a different and easier quantity: separating known pairs, which were already in the graph, from random pairs. Values near 0.99 are common in medical classification tasks, where each decision has two classes. In ranking thousands of drugs for an unseen disease, the field's best reported MRRs lie well below 1. Reviewers should be sceptical of near-perfect zero-shot numbers.

C. Limitations

(i) The targets have 1–24 labelled drugs, and PBC and AIH have one each, so per-target metrics are noisy and can be lower bounds (AIH). (ii) Validation diseases select the epoch; the selection-free test split (Section VII-D) bounds this bias. (iii) TxGNN's autoimmune area also contains autoinflammatory and non-specific nodes (e.g. familial Mediterranean fever, "autoimmune disease"), and the transfer test set E has only 12 diseases. The TxGNN comparison covers 13 test diseases of one split. (iv) Some ablations use one seed. (v) The model favours corticosteroids and antibiotics across immune diseases (41% and 21% of top-20 slots). This is clinically sensible for corticosteroids but partly popularity bias. (vi) PrimeKG labels are incomplete and partly derived from curated resources, so true repurposing candidates are scored as false positives. (vii) Literature labels were assigned with AI assistance and checked against abstracts, but have not been reviewed by a clinician. The temporal test covers only six drugs. (viii) No clinician has reviewed the candidates yet. The rankings are hypotheses for research, not treatment recommendations.

IX. Conclusion

Path-based zero-shot transfer works for rare autoimmune diseases that a model has never seen. The best source of the transferred knowledge depends on how much data is available. With only a handful of training diseases, related autoimmune diseases provide the most useful signal. With the full knowledge graph, autoimmune training data adds nothing measurable, and accuracy comes from broad pharmacological and biological structure. On a strict-family protocol with full ranking over all drugs, the path-based BioPathNet++ outperforms every baseline evaluated on the same split. It recovers established therapies for several rare autoimmune diseases on a single commodity GPU. Path reasoning drives accuracy, the transfer channel adds stability, and most of the graph's edges are redundant. An audit of an earlier

TABLE XIV
Highest-scoring explanation path for four predictions (main model, seed 0); edge labels are PrimeKG relations ($^{-1}$: inverse direction).

Pyridostigmine for Myasthenia Gravis: myasthenia gravis $\xrightarrow{\text{disease protein}^{-1}}$ MUSK $\xrightarrow{\text{disease protein}}$ neuromuscular junction disease $\xrightarrow{\text{indication (disease}\rightarrow\text{drug)}}$ Pyridostigmine
Ursodeoxycholic acid for Primary Biliary Cholangitis: primary biliary cholangitis $\xrightarrow{\text{disease disease}}$ genetic biliary tract disease $\xrightarrow{\text{disease disease}}$ choledocholithiasis $\xrightarrow{\text{indication (disease}\rightarrow\text{drug)}}$ Ursodeoxycholic acid
Amifampridine for Myasthenia Gravis: myasthenia gravis $\xrightarrow{\text{disease disease}}$ immune-mediated acquired neuromuscular junction disease $\xrightarrow{\text{disease disease}^{-1}}$ Lambert-Eaton myasthenic syndrome $\xrightarrow{\text{indication (disease}\rightarrow\text{drug)}}$ Amifampridine
Ciprofloxacin for Crohn Disease: Crohn disease $\xrightarrow{\text{disease disease}^{-1}}$ mouth disease $\xrightarrow{\text{disease disease}}$ maxillary sinusitis $\xrightarrow{\text{indication (disease}\rightarrow\text{drug)}}$ Ciprofloxacin

pipeline shows how leakage and random negatives can produce near-perfect but meaningless scores. Its unlabelled candidates are enriched for registered trials, and it ranks drugs approved after its training data above chance. On TxGNN's own split, trained only on edges that TxGNN also uses, it ranks all drugs significantly better than TxGNN. Future work: the TxGNN comparison on other disease areas, BioPathNet and KG-embedding baselines on a shared split, a three-seed test split, and clinician review of the candidates using the prepared review sheet.

Data and Code Availability

PrimeKG is available from Harvard Dataverse (doi:10.7910/DVN/IXA7BM). Code, splits, the audit script, run logs and the analysis that generates every number and table in this paper are in the project repository.

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